

EX:7 WRITE THE PROGRAM FOR DEPTH-FIRST SEARCH (DFS)

AIM

To develop a Python program that implements the **Depth-First Search (DFS)** algorithm to traverse all the nodes of a graph starting from a given source node.

ALGORITHM

Step 1: Start the program.

Step 2: Create the graph using an adjacency list.

Step 3: Accept the starting node from the user.

Step 4: Mark the starting node as visited and display it.

Step 5: Recursively visit each unvisited adjacent node until all reachable nodes are visited.

Step 6: End the program after traversing all the nodes.

PROGRAM CODE:

```
graph = {}
```

```
n = int(input("Enter number of vertices: "))
```

```
for i in range(n):
```

```
    node = input("Enter vertex: ")
```

```
    neighbours = input("Enter neighbours separated by space: ").split()
```

```
    graph[node] = neighbours
```

```
start = input("Enter starting node: ")
```

```
visited = set()
```

```

def dfs(node):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

    for neighbour in graph[node]:
        dfs(neighbour)

print("\nDFS Traversal:")
dfs(start)

```

OUTPUT:

```

===== RESTART: C:/Users/Hp/AppData/Local/Programs/Py
Enter number of vertices: 4
Enter vertex: A
Enter neighbours separated by space: B C
Enter vertex: B
Enter neighbours separated by space: D
Enter vertex: C
Enter neighbours separated by space:
Enter vertex: D
Enter neighbours separated by space:
Enter starting node: A

Graph:
  A
 / \
B   C
 \ /
  D

DFS Traversal:
A B D C
|

```

RESULT:

The Python program successfully implements the **Depth-First Search (DFS)** algorithm to traverse the graph. The nodes are visited and displayed in **depth-first order** starting from the given source node.